

Assortative pairing raises IBD F in a closed vial of 1,000 flies

A note on the fly_vial seed-1 lock. Not a result. Locks stay on the trees.

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Index of locks: <https://gist.github.com/martialsystems/12835f747d6360781f3cc7f91f243178>

Locks stay on the trees. Index: gist 12835f74.

Abstract

Question: does a closed vial of 1,000 diploid flies, paired by genome similarity, raise IBD F faster than random mating at the same N, seed, and load?

Yes on the seed-1 lock. $k=3$, $N=1,000$, 80 generations: within-individual IBD $F = 0.524$ versus random $F = 0.034$. Census held. Courtship fell on both arms. Engine is closed-form. FlyWire and MaleCNS are templates, not the stepper. Science lock @e2e22b7. Logs: logs/assort_80.json, logs/random_80.json. Do not restamp 0.524 / 0.034.

Seeds 2 and 3 are a screen, not an effect-size estimate. Assortative $F = 0.474$ and 0.463 . Random $F = 0.035$ and 0.035 . All three assortative arms sit well above Wright's random-mating scale. $n = 3$ does not promote the row from Screen to an estimate.

The question was locked after the split (pre_specified: false in METHODS.yaml). Falsifier for the seed-1 lock: assortative F not greater than random F at the same N, seed, and load.

Methods card copied from fly_vial METHODS.yaml:

Field	Value
Object	unconstrained evolutionary toy
Status	Screen
System	$N=1,000$; 80 generations; $k=3$ vs random
Dynamics	closed-form QTL scalars; no LIF
RNG	<code>numpy.random.default_rng(seed)</code>
n / seeds	3 (seed 1 lock; seeds 2 and 3 audit files)
Locked metric	assortative $t=80$ $F=0.524$ vs random $F=0.034$
Science lock	e2e22b7

A second locked split on the same engine asked whether $k=3$ drives failure faster when census is allowed to move. --cap off still clipped from above. Realized viable births stayed above 1,000 on every generation. T_fail never fired. Collapse under similarity pairing remains untested.

Keywords

Drosophila; assortative mating; IBD; Wright inbreeding; closed vial; FlyWire; MaleCNS.

1. Question

Does a closed vial of 1,000 diploid flies, paired by genome similarity, raise IBD F faster than random mating at the same N, seed, and load?

Object of study: unconstrained evolutionary toy. Connectome templates label circuit scalars. They do not step the vial. Status on the index row is Screen because $n = 3$. The seed-1 split is closed as a Yes.

2. Falsifier and protocol

Falsifier (seed-1 lock): assortative F not greater than random F at the same N, seed, and load.

Protocol copied from the locked JSON. N = 1,000 diploid adults. 80 non-overlapping generations. Mating is forced k=3 nearest-neighbor pairing on morphology plus courtship-circuit QTLs, or random. Hidden recessive load (16 lethals, 48 sublethals, h = 0) is excluded from similarity. Offspring genomes are Mendelian plus per-locus Gaussian mutation. Density cap samples viable adults uniformly down to 1,000. Recombination is free: each locus segregates independently.

F in the log is $1 - H_t / H_0$ from within-individual founder-allele IBD. $H_0 = 1$, so $H = 1 - F$ is the same number written twice. Heterozygosity is IBD on founder IDs, not a floating-point compare of QTL values. `mean_pairwise_phi` is mean kinship over unordered adult pairs. It is not F (assortative t = 80: F = 0.524, phi = 0.076).

RNG is `numpy.random.default_rng(seed)`. The eval-connectome hook uses `default_rng(seed + 1_000_003)` and does not step the vial. Seed-1 locked logs are hook-off. Seeds 2 and 3 may turn the hook on as an audit. They write new files. They do not replace seed 1.

Population-genetics prior: Wright's exact random-mating inbreeding $1 - (1 - 1/2N)^t$ is the scale for the random arm (Wright, 1931, Genetics 16:97-159, doi 10.1093/genetics/16.2.97). At N = 1,000, t = 80 that form is about 0.039. The linear headline $80 / 2,000 = 0.04$ is that scale. Logged random F = 0.034 sits a little under it. Assortative k-NN is the treatment.

3. Locked seed-1 table

Copied from `logs/assort_80.json` and `logs/random_80.json`. Courtship at t = 0 is omitted: founders have not paired.

Assortative:

t	n	F	egg viability	fertility	courtship	accepted pairs	clusters
--:	--:	--:	-----:	-----:	-----:	-----:	-----:
0	1,000	0.000	0.963	0.997		0	4
1	1,000	0.000	0.921	0.991	0.934	372	32
8	1,000	0.131	0.817	0.966	0.820	385	98
20	1,000	0.217	0.787	0.959	0.744	374	75
40	1,000	0.399	0.777	0.952	0.680	382	61
80	1,000	0.524	0.748	0.958	0.595	386	48

Random:

t	n	F	egg viability	fertility	courtship	accepted pairs	clusters
--:	--:	--:	-----:	-----:	-----:	-----:	-----:
0	1,000	0.000	0.963	0.997		0	4
1	1,000	0.000	0.918	0.990	0.930	493	10
8	1,000	0.005	0.871	0.979	0.861	493	77
20	1,000	0.008	0.832	0.969	0.772	488	25
40	1,000	0.021	0.784	0.959	0.690	481	7
80	1,000	0.034	0.791	0.959	0.599	485	2

Independent fitness numbers, egg-to-adult viability, t = 0 to t = 80: assortative 0.963 to 0.748; random 0.963 to 0.791. Fertility: 0.997 to 0.958 assortative, 0.997 to 0.959 random. Courtship uses the first generation that pairs (t = 1), then t = 80: 0.934 to 0.595 assortative, 0.930 to 0.599 random. Those two courtship series move together. The courtship drop is not the assortative signature.

4. Three-seed screen

Seeds 2 and 3 write `logs/assort_80_s2.json` and friends. Assortative $F = 0.474$ (seed 2) and 0.463 (seed 3), both much greater than Wright (~ 0.04). Random $F = 0.035$ and 0.035 . Courtship still falls on both arms. Eval hook was on for those four runs only, as an audit. Title stays seed 1.

$n = 3$ is a screen. A later tree may rerun the locked protocol at $n \geq 10$ and report the distribution. That run would write new files. It would not restamp `logs/assort_80.json`.

5. Cap-off split

Question: at the same seed, load, and k , does $k=3$ similarity pairing drive the vial to failure faster than random mating when census is allowed to move?

This run did not answer that. `--cap off` only refuses to invent adults when short. The 1,000 ceiling still clips from above. Realized viable births (`n_viable` before clip) stayed above 1,000 on every generation of every vial. Display `n` was pinned at 1,000 from the surplus. `T_fail` rules ($n = 0$, accepted pairs = 0, or $n < 50$ and egg viability < 0.20) never fired because they cannot fire while that surplus holds. Removing the cap did not make similarity pairing the faster route to failure.

seed	knn T_fail	random T_fail	knn min n_viable	random min n_viable
1	200	200	1,959	2,644
2	200	200	1,997	2,509
3	200	200	1,970	2,464

Two locked results, zero collapse results: capped $k=3$ raises F with census held and courtship falling on both arms; ceiling still binding, nobody dies by $t=200$.

6. Reduction

Female template: FlyWire 139,255 neurons, whole brain, no VNC (Dorkenwald et al., Nature 2024, doi 10.1038/s41586-024-07558-y). Male template: MaleCNS 166,691 neurons, brain plus ventral nerve cord (Berg et al., Cell 189:5504-5526, 2026-09-03, doi 10.1016/j.cell.2026.08.015). The tree locks 166,691. Female locomotion and copulation-motor scores are closed-form overlays on that brain map. `vpoDN` and song CPG names in the G2P map are circuit labels for those scalars.

Original `fly_vial` code is MIT. MaleCNS is CC BY 4.0. A citation file (`CITATION.cff`) lives on the tree. Artifact DOI is empty until a Zenodo (or other) deposit exists. Do not invent a DOI.

This note does not fold the vampire trees. Those questions live in `vial_vampire_writeup` and halt at 0 of 3 on bite share and mosquito theft. The closed-vial block on the index gist states the same template rule for the whole arm.

7. Replay

Hash-check or rebuild in a temp path. Never overwrite the lock files.

```
.venv/bin/python -m pytest
.venv/bin/python scripts/reproduce_lock.py
.venv/bin/python scripts/reproduce_lock.py --rerun
```

Those commands run in `fly_vial @e2e22b7`. Pin: `numpy 2.5.3`, `pytest 9.1.1`, `numpy.random.default_rng`. Expected sha256 of `logs/assort_80.json`:
92a328278e62c4c607749c9358cdc733e1f06424a4b7f13c97c352ce4f4f759b.

Accountability (2026-09-20)

Results. Seed-1 metrics are Python engine output at tag science-e2e22b7 (logs/assort_80.json, logs/random_80.json; display F = 0.524 vs 0.034). They were not generated by a language model. The drafting tool did not touch the engine.

Text. The computational note (NOTE.md / docs/fly_vial_f_note.pdf), the methods card, and the index gist prose were drafted with Grok (xAI). The depositor reviewed the text against the locked logs and takes responsibility for it. No AI tool is an author.

Release: https://github.com/martialsystems/fly_vial/releases/tag/science-e2e22b7

Sources (2026-09-20)

Records checked on Crossref, 2026-09-20.

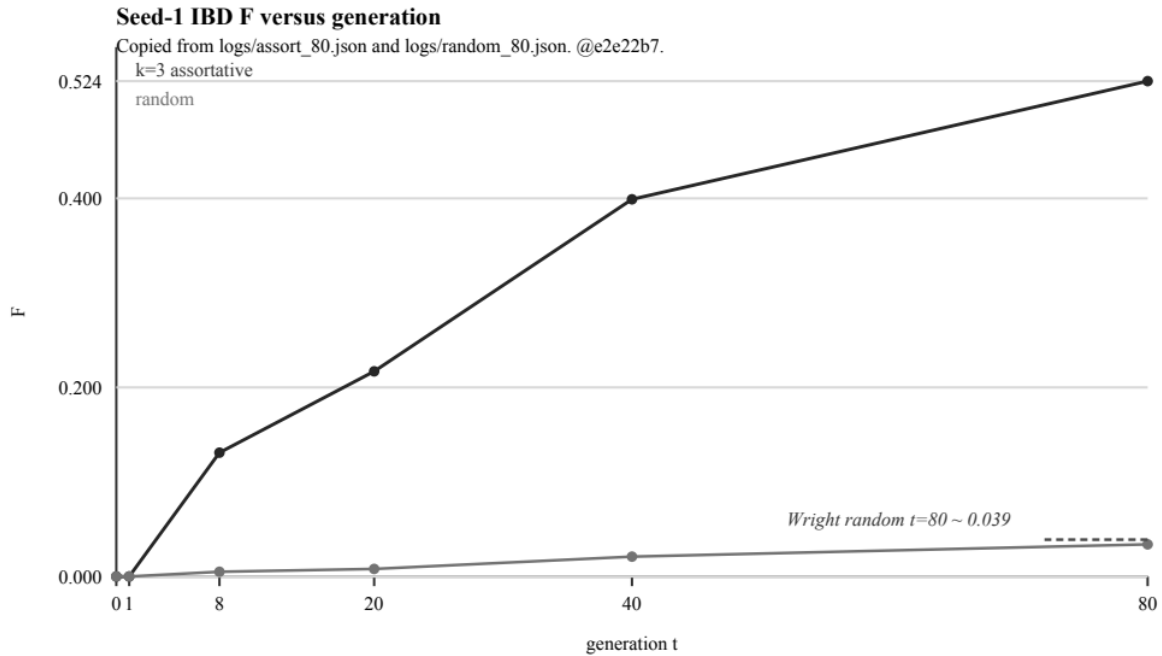
Berg, S., Beckett, I. R., Costa, M., Schlegel, P., Januszewski, M., Marin, E. C., Nern, A., et al. (2026). Sexual dimorphism in the complete *Drosophila* male central nervous system connectome. *Cell*, 189(18), 5504-5526.e15. <https://doi.org/10.1016/j.cell.2026.08.015>

Dorkenwald, S., Matsliah, A., Sterling, A. R., Schlegel, P., Yu, S. C., McKellar, C. E., Lin, A., et al. (2024). Neuronal wiring diagram of an adult brain. *Nature*, 634(8032), 124-138. <https://doi.org/10.1038/s41586-024-07558-y>

Wright, S. (1931). Evolution in Mendelian populations. *Genetics*, 16(2), 97-159. <https://doi.org/10.1093/genetics/16.2.97>

Figure 1 (figures/stack.png): F versus generation on the seed-1 lock, assortative versus random, with Wright's random-mating form at t = 80 as a scale mark.

Index (pointers only): <https://gist.github.com/martialsystems/12835f747d6360781f3cc7f91f243178>

Figure 1. Seed-1 IBD F versus generation (2026-09-19).

Assortative $k=3$ versus random mating. Wright random-mating form at $t=80$ is a scale mark, not a fit. Copied from the locked JSON.

Revisions

2026-09-19: first methods note from the locked seed-1 logs and the three-seed screen. Vampire trees stay in vial_vampire_writeup.

2026-09-20: sources (Wright 1931, Dorkenwald 2024, Berg 2026); accountability sentence. Release pointer. No DOI.